MLOps Group Project Emotion Classification Pipeline

End-to-End Report

Group 2 PGD AI IIT Jodhpur

| Name | Roll Number | Primary Contribution | |------|-------------|---------------------| | Nikhil Saini | G25AIT2067 | Repo setup, branch protection, CI/CD workflows (ci.yml, inference.yml, docker.yml), GitHub Secrets, report compilation | | Y Sharathchandrika | G25AIT2132 | Data inspection, cleaning pipeline (`data.py`), `id2label.json`, class-distribution analysis | | Sarthak Kapoor | G25AIT2098 | Model training (2 Kaggle versions), W&B tracking, HF model push, W&B dashboard public setup | | Aryaveer Rathi | G25AIT2021 | `inference.py`, Dockerfile design, local Docker build/test, inference workflow support |

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1. Project Overview

We built a complete MLOps pipeline for **6-class text emotion classification** using the `dair-ai/emotion` dataset. Two transformer models were trained with different architectures and hyperparameters on Kaggle GPU, tracked with Weights & Biases, and the best model was containerised with Docker and deployed via GitHub Actions.

Dataset: `dair-ai/emotion` (Hugging Face) - 16,000 train / 2,000 validation / 2,000 test - Classes: sadness, joy, love, anger, fear, surprise

All public links: | Resource | URL | |----------|-----| | GitHub Repository | https://github.com/nikhilsaini-iitj/MLOps_GroupProject | | Kaggle Notebook v1 (MiniLM) | https://www.kaggle.com/code/nikhilg25ait2067/ | | Kaggle Notebook v2 (ELECTRA) | https://www.kaggle.com/code/nikhilg25ait2067/ | | Hugging Face Model v1 | https://huggingface.co/Nikhil-iitj/emotion-minilm | | Hugging Face Model v2 (winner) | https://huggingface.co/Nikhil-iitj/emotion-electra | | W&B Dashboard | https://wandb.ai/g25ait2067-prom-iit-rajasthan/mlops-groupproject-v2 | | Docker Image | https://hub.docker.com/r/nikhilsainiiitj/mlops-groupproject-inference |

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2. GitHub Repository Setup (Task 1 10 marks)

Owner: Nikhil Saini

We created a public repository `MLOps_GroupProject` under the organisation account. The repository uses a `main` + `develop` branching strategy: - `main`: production-ready code, protected by branch rules - `develop`: integration branch for all feature work

Branch protection on `main`: - Require a pull request before merging - Require 1 approving review - Dismiss stale PR approvals when new commits are pushed

All four team members were added as collaborators with **Write** access. The `.github/workflows/` directory contains three workflows (see Tasks 6 & 7).

Screenshot: Collaborators page saved in submission folder.

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3. Data Preparation (Task 2 15 marks)

Owner: Y Sharathchandrika

The `dair-ai/emotion` dataset was loaded via `datasets.load_dataset()`. Preprocessing steps applied in `src/data.py`:

1. **Lowercasing** standardises text casing
2. **Punctuation stripping** removes non-alphanumeric characters (except apostrophes)
3. **Whitespace collapse** replaces multiple spaces with single space
4. **Null handling** verified no null samples exist
5. **Duplicate removal** dropped exact duplicate text rows
6. **Label encoding** generated `id2label.json` and `label2id` mapping: `json {"0": "sadness", "1": "joy", "2": "love", "3": "anger", "4": "fear", "5": "surprise"}`

Class distribution note: The dataset is mildly imbalanced `joy` (~30%) and `sadness` (~28%) dominate, while `surprise` (~8%) and `love` (~14%) are under-represented. We documented this rather than rebalancing, as the rubric asks for inspection and rationale, not silent fixes.

Commit: `id2label.json` is tracked; raw/processed data is `.gitignored`.

4. Model Selection (Task 3 10 marks)

Owner: Sarthak Kapoor

Both models were selected from the Hugging Face Hub with the constraint **< 200 MB** on disk.

v1: `microsoft/MiniLM-L12-H384-uncased`

- **Architecture:** 12-layer, 384-hidden distilled transformer (Wang et al., 2020)
- **Objective:** General-domain distillation from BERT-base
- **Parameters:** ~33 M
- **On-disk size:** ~133.5 MB (`model.safetensors`)
- **Rationale:** Deep but narrow; strong baseline for short-text classification

v2: `google/electra-small-discriminator`

- **Architecture:** Small ELECTRA with replaced-token detection (Clark et al., 2020)
- **Objective:** RTD (predict which tokens were replaced) rather than MLM
- **Parameters:** ~14 M
- **On-disk size:** ~54.2 MB (`model.safetensors`)
- **Rationale:** More sample-efficient pretraining objective; fewer params but competitive downstream performance

Both models were loaded with `AutoModelForSequenceClassification`, wiring `id2label`/`label2id` so the classifier head outputs human-readable labels.

5. Training Two Versions (Task 4 25 marks)

Owner: Sarthak Kapoor

Training was executed on **Kaggle GPU T4** using the Hugging Face `Trainer` API. Secrets (`WANDB_API_KEY`, `HF_TOKEN`) were stored in **Kaggle Secrets** never hardcoded.

Hyperparameters

Parameter	v1 (MiniLM)	v2 (ELECTRA-small)
Epochs	3	4
Batch size	16	16
Learning rate	3e-5	5e-5
Max sequence length	128	128
Optimiser	AdamW	AdamW
Weight decay	0.01	0.01
Eval strategy	epoch	epoch
Save strategy	epoch	epoch
Best metric	F1 (weighted)	F1 (weighted)

Validation Results (epoch-wise)

v1 (MiniLM):

Epoch	Train Loss	Val Loss	Accuracy	F1
1	1.203	1.089	0.812	0.798
2	0.612	0.589	0.901	0.901
3	0.341	0.542	0.917	0.917

v2 (ELECTRA-small): | Epoch | Train Loss | Val Loss | Accuracy | F1 | |-------|-----------|----------|----------|-----| | 1 | 1.181 | 1.068 | 0.846 | 0.832 | | 2 | 0.565 | 0.560 | 0.919 | 0.919 | | 3 | 0.365 | 0.428 | 0.929 | 0.930 | | 4 | 0.295 | 0.393 | 0.931 | 0.931 |

Final Test Metrics

| Version | Model | Test Accuracy | Test F1 | Test Loss | Size | Runtime | |---------|-------|--------------|---------|-----------|------|---------| | v1 | MiniLM-L12 | 0.907 | 0.907 | 0.542 | 133.5 MB | ~3m 44s | | **v2** | **ELECTRA-small** | **0.9265** | **0.9269** | **0.395** | **54.2 MB** | **~2m 56s** |

Winner: v2 (ELECTRA-small) higher accuracy, lower loss, **2.5 smaller**, and faster inference. The RTD pretraining objective appears to produce more efficient representations for this downstream task.

W&B tracking: `report_to='wandb'`, `logging_strategy='steps'`, `logging_steps=20`. We used a fresh project (`mlops-groupproject-v2`) with explicit `define_metric` calls to ensure charts render correctly against epoch/step axes.

6. Hugging Face Model Push (Task 5 5 marks)

Owner: Sarthak Kapoor

The winning v2 model (best checkpoint selected by `load_best_model_at_end=True` with `metric_for_best_model='f1'`) was pushed to Hugging Face Hub:

```
python trainer.save_model('best') tok.save_pretrained('best')
model.push_to_hub('Nikhil-iitj/emotion-electra')
tok.push_to_hub('Nikhil-iitj/emotion-electra')
```

- **Repository:** https://huggingface.co/Nikhil-iitj/emotion-electra
- **Visibility:** Public
- **Contents:** `model.safetensors` (54.2 MB), `config.json`, `tokenizer.json`, `vocab.txt`, `id2label` mapping
- **W&B link:** `wandb.run.summary['huggingface_model']` points to the same URL

7. Docker Container (Task 6 10 marks)

Owner: Aryaveer Rathi

Dockerfile Design

```
dockerfile FROM python:3.11-slim ARG HF_MODEL_NAME=Nikhil-iitj/emotion-electra
ENV HF_MODEL_NAME=${HF_MODEL_NAME} WORKDIR /app COPY requirements.txt . RUN pip
install --no-cache-dir -r requirements.txt COPY src/ ./src/ COPY id2label.json .
ENTRYPOINT ["python", "src/inference.py"]
```

Design decisions: - **Slim base:** `python:3.11-slim` keeps image small (~180 MB final) - **Build-arg:** `HF_MODEL_NAME` allows switching models without editing source - **Minimal deps:** `requirements.txt` includes only `torch`, `transformers`, `huggingface_hub` for inference (training deps excluded) - **Runtime token:** `HF_TOKEN` passed as env var at `docker run`, never baked into image

Local Build & Test

```bash docker build --build-arg HF_MODEL_NAME=Nikhil-iitj/emotion-electra \ -t mlops-groupproject-inference:latest .

docker run --rm -e HF_TOKEN=$HF_TOKEN \ -e INPUT_TEXT="I am so happy today!" \ mlops-groupproject-inference:latest ```

**Output:** `Input : I am so happy today! Label : joy (confidence 0.987)`
```

Docker Hub Push

```
bash docker tag mlops-groupproject-inference:latest \
nikhilsainiiitj/mlops-groupproject-inference:latest docker push
nikhilsainiiitj/mlops-groupproject-inference:latest
```

- **Image URL:** https://hub.docker.com/r/nikhilsainiiitj/mlops-groupproject-inference
- **Visibility:** Public
- **Tag:** `latest` (also v2 for version pinning)

Automated workflow: `.github/workflows/docker.yml` builds and pushes on every push to `main`.

8. GitHub Actions (Task 7 15 marks)

Owner: Nikhil Saini

Workflow 1: CI (`.github/workflows/ci.yml`)

- **Trigger:** Push to `develop`, PR to `main`
- **Job:** `flake8 src/ --max-line-length=120`
- **Purpose:** Code quality gate before merge

Workflow 2: Inference (`.github/workflows/inference.yml`)

- **Trigger:** `workflow_dispatch` (manual button in GitHub UI)
- **Inputs:**
- `input_text`: text to classify
- `hf_model_name`: defaults to `Nikhil-iitj/emotion-electra`
- **Job:** Installs deps, sets `HF_TOKEN` from repo Secrets, runs `src/inference.py`
- **Secrets used:** `HF_TOKEN`

Workflow 3: Docker Build & Push (`.github/workflows/docker.yml`)

- **Trigger:** Push to `main`, manual dispatch
- **Job:** Buildx login to Docker Hub push `latest` + `v2` tags
- **Secrets used:** `DOCKERHUB_USERNAME`, `DOCKERHUB_TOKEN`

Successful run evidence: Screenshots of Actions runs + logs saved in submission folder.

9. Weights & Biases Dashboard (Task 8 5 marks)

Owner: Sarthak Kapoor

- **Project:** `mlops-groupproject-v2`
- **URL:** https://wandb.ai/g25ait2067-prom-iit-rajasthan/mlops-groupproject-v2
- **Visibility:** Public
- **Runs:** `run-v1` (MiniLM) and `run-v2` (ELECTRA-small)
- **Metrics logged:** `train/loss`, `eval/loss`, `eval/accuracy`, `eval/f1`, `train/epoch`, `train/global_step`, hyperparameters
- **Step configuration:** `train/*` plotted against `train/global_step`; `eval/*` plotted against `epoch`

Comparison view: The Workspace panel shows both runs side-by-side, confirming v2 outperforms v1 on accuracy, F1, and loss while using a smaller model.

Screenshot: W&B Workspace with both runs inserted in Section 4.2.1 of this report.

10. Link Verification (Incognito Audit)

All five required links were opened in an incognito/private window before submission:

| # | Link | Status | |---|------|--------| | 1 | GitHub repo | Public, workflows visible | | 2 | Kaggle v1 notebook | Public | | 3 | Kaggle v2 notebook | Public | | 4 | HF model (v2 winner) | Public, files downloadable | | 5 | W&B dashboard | Public, charts render | | 6 | Docker Hub image | Public, pullable |

11. Challenges & Learnings

Model size trap: Our initial choice (`distilbert-base-uncased`, ~268 MB) exceeded the 200 MB limit. We pivoted to MiniLM (~120 MB) and ELECTRA-small (~54 MB) after reviewing on-disk `model.safetensors` sizes.

W&B blank charts: Early runs showed "There's no data for the selected runs." Root cause: `Trainer` logs eval metrics sparsely (only at epoch boundaries) but W&B default panels expect continuous step data. **Fix:** added `logging_strategy='steps'`, `logging_steps=20`, and explicit `define_metric` calls mapping `eval/*` to epoch axis.

HF_TOKEN whitespace: Kaggle Secrets appended trailing whitespace causing `LocalProtocolError` on Hugging Face login. **Fix:** `.strip()` on `secrets.get_secret('HF_TOKEN').strip()`.

Kaggle Internet toggle: `UserSecretsClient` fails silently if Kaggle Settings Internet is OFF. Documented prominently in notebook headers.

GitHub contributor hygiene: Early commits included `Co-Authored-By: Claude` trailers, causing `@claude` to appear in contributor graphs. **Fix:** amended commits with `git commit --amend` removing trailers; recreated repo when cached data persisted.

12. Repository Structure

```
MLOps_GroupProject/ .github/workflows/ ci.yml # Lint on develop push / PR to main inference.yml # Manual inference trigger docker.yml # Auto-build & push on main src/ data.py # Data loading + cleaning train.py # Local training template eval.py # Evaluation + artifact logging utils.py # compute_metrics, TextDataset, load_label_map inference.py # Docker / Actions entrypoint kaggle_minilm.ipynb # Kaggle notebook v1 kaggle_electra.ipynb # Kaggle notebook v2 Dockerfile # Inference container requirements.txt # Python dependencies id2label.json # Label mapping (committed) .gitignore # Excludes data/, results/, .env README.md # Setup + all live links TEAM_DECISION.md # Model-selection voting doc PLAN_v2.md # Execution plan with role assignments REPORT.md # This file
```

Submitted by Group 2, IIT Jodhpur